Columbia Econ Math Camp 2025

Correspondences

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Correspondences

Definition 1 (Correspondences)

A **correspondence** F from X to Y is a set valued function that associates every element in X to a subset of Y, denoted $F : X \rightrightarrows Y$ or $x \mapsto F(x)$. The set X is called the **domain** of F, Y is the **codomain**, and $F(x)$ is the **image** of F at $x \in X$. If $U \subset X$, we also denote $F(U) := \bigcup_{x \in U} F(x)$.

- A correspondence is just a function that maps elements of the domain onto *subsets of the codomain*.
- Alternatively, a function is just a single-valued correspondence - i.e. any image is a set with one element.
- For notation purposes, if every image of F (or F at $x_0 \in X$) satisfies a property P, we say F is P-valued (or at x_0).
- For example, property P can be non-empty-, single- (or singleton-), open-, closed-, compact-, and/or convex-valued.

Upper Hemicontinuity

Continuity in correspondences is more nuanced since images can move, shrink or enlarge.

Definition 2 (Upper hemicontinuity)

Let $(X, d_X), (Y, d_Y)$ be metric spaces. We say $F : X \rightrightarrows Y$ is **upper hemicontinuous (or uhc) at** $x_0 \in X$ iff for any open set $U \subset Y$ that contains $F(x_0)$, there exists an open ball around x_0 whose image is contained in U, i.e. $\exists \delta > 0$ s.t. $F(x) \subset U$ for any $x \in B_\delta(x_0)$. If F is uhc at every $x_0 \in X$ we say F is **upper hemi-continuous (uhc)**.

- Visually, F is uhc at x_0 if for any open set that covers $F(x_0)$, then *images of points nearby* x_0 are also covered by the same open set.
- Basically, this precludes images from suddenly or *discontinuously* growing too big, or *enlarging*.

Upper Hemicontinuity

The idea can be seen clearly from the following example:

Exercise 1

1. Consider $F_1 : \mathbb{R} \rightrightarrows \mathbb{R}$ with

$$F_1(x) = \begin{cases} \{0\}, & \text{if } x < 0 \\ [-1, 1], & \text{otherwise} \end{cases}$$

   Show that F_1 is uhc at 0.

2. Consider $F_2 : \mathbb{R} \rightrightarrows \mathbb{R}$ with

$$F_2(x) = \begin{cases} \{0\}, & \text{if } x \leq 0 \\ [-1, 1], & \text{otherwise} \end{cases}$$

   Show that F_2 is not uhc at 0.

Upper Hemicontinuity

More examples about the nuances of uhc

Exercise 2

1. Show that $F_3 : \mathbb{R} \rightrightarrows \mathbb{R}$ with $F_3(x) = [x, x+1]$ is uhc.
2. Show that $F_4 : \mathbb{R} \rightrightarrows \mathbb{R}$ with $F_4(x) = (x, x+1)$ is not uhc.

- F_4 is not uhc because the definition allows $U = F(x_0)$ and naturally no other image outside of x_0 will contain U.
- For the most part we will deal with closed-valued correspondences, so this should not be an issue.

Upper Hemicontinuity

Uhc has a sequence-based definition that is simpler to manipulate:

Theorem 1 (Sequential characterization of uhc)

Let $(X, d_X), (Y, d_Y)$ be metric spaces, $F : X \rightrightarrows Y$, $x_0 \in X$. The following two statements are equivalent:

1. F is uhc and compact-valued at x_0.
2. For any sequence (x_n, y_n) in $X \times Y$ where $x_n \to x_0$ and $y_n \in F(x_n)$ for any $n \in \mathbb{N}$, there exists some subsequence (y_{n_k}) and $y_0 \in F(x_0)$ s.t. $y_{n_k} \to y_0$.

- This only works when F is compact-valued! See F_4 from previous slide to see why this characterization fails when non compact-valued.

Upper Hemicontinuity

Finally, uhc generalizes continuity for functions:

Exercise 3

Show that a function $f : X \to Y$ is continuous iff $F : X \rightrightarrows Y$ with $F(x) := \{f(x)\}$ is uhc.

Lower Hemicontinuity

Another related notion of continuity for correspondences:

Definition 3 (Lower hemicontinuity)

Let $(X, d_X), (Y, d_Y)$ be metric spaces. We say $F : X \rightrightarrows Y$ is **lower hemicontinuous at** $x_0 \in X$ iff for any open set $U \subset Y$ that has some non-empty intersection with $F(x_0)$ - i.e. $F(x_0) \cap U \neq \emptyset$ - there exists some open ball around x_0 where every images each has a non-empty intersection with U, i.e. $\exists \delta > 0$ s.t. $F(x) \cap U \neq \emptyset$ for any $x \in B_\delta(x_0)$.

- Visually, if an open set has common ground with $F(x_0)$, then the *images of points nearby* x_0 must also have some common ground.
- This one now precludes images of nearby points of x_0 from discontinuously becoming too small compared to $F(x_0)$, or *shrinking*.

Lower Hemicontinuity

Let's look at the same examples we developed for uhc:

Exercise 4

1. Consider $F_1 : \mathbb{R} \rightrightarrows \mathbb{R}$ with
$$F_1(x) = \begin{cases} \{0\}, & \text{if } x < 0 \\ [-1, 1], & \text{otherwise} \end{cases}$$
Show that F_1 is not lhc at 0.
2. Consider $F_2 : \mathbb{R} \rightrightarrows \mathbb{R}$ with
$$F_2(x) = \begin{cases} \{0\}, & \text{if } x \leq 0 \\ [-1, 1], & \text{otherwise} \end{cases}$$
Show that F_2 is lhc at 0.

Lower Hemicontinuity

Again let's revisit these uhc examples:

Exercise 5

1. Show that $F_3 : \mathbb{R} \rightrightarrows \mathbb{R}$ with $F_3(x) = [x, x+1]$ is lhc.
2. Show that $F_4 : \mathbb{R} \rightrightarrows \mathbb{R}$ with $F_4(x) = (x, x+1)$ is lhc.

- Note that F_4 is now lhc!

Lower Hemicontinuity

- As you can begin to see, both definitions are linked.
- To see how, for a given $F : X \rightrightarrows Y$, define $F^c : X \rightrightarrows Y$ as $F^c(x) = Y - F(x)$.
- The following proposition establishes the link.

Proposition 1

Let $(X, d_X), (Y, d_Y)$ be metric spaces, $F : X \rightrightarrows Y$, $x_0 \in X$. If F is uhc at x_0, then F^c is lhc at x_0.

Lower Hemicontinuity

Lhc has a more flexible sequential definition too

Theorem 2 (Sequential characterization of lhc)

Let $(X, d_X), (Y, d_Y)$ be metric spaces, $F : X \rightrightarrows Y$, $x_0 \in X$. The following are equivalent:

1. F is lhc at x_0.
2. For any $y_0 \in F(x_0)$ and (x_n) in X with $x_n \to x_0$, there exists some (x_{n_k}) and (y_k) s.t. $y_k \in F(x_{n_k})$ and $y_k \to y_0$.

Lower Hemicontinuity

As with uhc, lhc generalize continuity of functions

Exercise 6

Show that a function $f : X \to Y$ is continuous iff $F : X \rightrightarrows Y$ with $F(x) := \{f(x)\}$ is lhc.

Because they are both complements of each other, we can define continuity of a correspondence as follows

Definition 4 (Continuity of correspondences)

Let (X, dX), (Y, dY) be metric spaces. We say $F : X \rightrightarrows Y$ is said to be **continuous at** $x_0 \in X$ iff F is both uhc and lhc at x_0. Moreover F is said to be **continuous** iff F is continuous at every point in its domain.

Graphs

Let's define the notion of a closed graph which is tightly related to uhc

Definition 5 (Closed graph)

Let $(X, d_X), (Y, d_Y)$ be metric spaces. We say $F : X \rightrightarrows Y$ has the **closed graph property (cgp) at** $x_0 \in X$ if for any $(x_n, y_n) \in X \times Y$ with $x_n \to x_0$, $y_n \in F(x_n)$ and $y_n \to y_0 \in Y$, it holds that $y_0 \in F(x_0)$. Moreover, if this holds in every point of its domain, we say F has the **closed graph property (cgp)** or has a **closed graph**.

Exercise 7

Show that F is cgp then F is closed-valued. Show that the reverse may not hold.

Graphs

- Cgp in practice just means that the set made up of the graph of F is itself a closed set.
- Specifically, for metric spaces $(X, d_X), (Y, d_Y)$ and correspondence $F : X \rightrightarrows Y$, define the **graph** of F as

$$Gr(F) := \{(x, y) \in X \times Y : y \in F(x)\}$$

and $d_{X\times Y}((x, y), (x', y')) := \sqrt{d_X(x, x')^2 + d_Y(y, y')^2}$, which you can show is a valid metric function.

Proposition 2

Let $(X, d_X), (Y, d_Y)$ be metric spaces. Then F has closed graph iff $Gr(F)$ is closed in $(X \times Y, d_{X\times Y})$.

Graphs

Definition 6 (Local boundedness)

Let $(X, d_X), (Y, d_Y)$ be metric spaces. We say $F : X \rightrightarrows Y$ is **locally bounded at** $x_0 \in X$ if there exists $\delta > 0$ and a compact set K in Y such that $F(B_\delta(x_0)) \subset K$.

With this, the link between cgp and uhc is given by the following proposition

Proposition 3

Let $(X, d_X), (Y, d_Y)$ be metric spaces, $F : X \rightrightarrows Y$ is closed-valued and locally bounded at $x_0 \in X$. Then following two statements are equivalent:

1. F is uhc at x_0.
2. F is cgp at x_0.

Important Theorems**

These notions are useful to derive the Kakutani FPT, which is useful to prove existence of Nash equilibriums and also Walrasian equilibrium in general equilibrium theory:

Theorem 3 (Kakutani's Fixed Point)

Let X be non-empty, compact, convex set in $\mathbb{R}^n$. If $F : X \rightrightarrows X$ is nonempty-valued, compact-valued, convex-valued and uhc, then there exists a fixed point of F, i.e. some $x^* \in X$ s.t. $x^* \in F(x^*)$.

- Note that if F is single-valued, this just becomes Brouwer's FPT.
- Note that since X is compact, we can replace *compact-valued* $+$ *uhc* $=$ *cgp* in the theorem.

Important Theorems**

Finally we have Berge's Theorem of the Maximum:

Theorem 4 (Berge's Theorem of the Maximum)

Let $(X, d_X), (A, d_A)$ be metric spaces, $f : X \times A \to \mathbb{R}$ is continuous the metric $d_{X\times A}$, $\alpha_0 \in A$, and $D : A \rightrightarrows X$ at α_0 is nonempty-valued, compact-valued and continuous. Define $X^* : A \rightrightarrows X$ as

$$X^*(\alpha) := \arg\max_{x\in X} \{f(x,\alpha) : x \in D(\alpha)\}$$

Let $\hat{A} := \{\alpha \in A : X^*(\alpha) \neq \emptyset\}$, and define $f^* : \hat{A} \to \mathbb{R}$ as

$$f^*(\alpha) := \max_{x\in X} \{f(x,\alpha) : x \in D(\alpha)\}$$

Then X^* at α_0 is nonempty-valued, compact-valued and uhc, and f^* is continuous at α_0.